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DBMS Assignment

Section - A

Q1 = What is Database Management System (DBMS)? Why is it required?

Ans = DBMS is a software that enables users to define, create, maintain and control access to database
It is required because it controls data redundancy, maintains data consistency, security and ensures data integrity.

Q2 = Why is DBMS better than traditional file system?

Ans = DBMS used centralized controls, whereas file systems store data seperately in files. Reduced redundancy, controlled consistency, access controls, backup and recovery.

Q3 = Data Abstraction and its levels

Ans = Data Abstractions hides internal storage details from users.
Three levels :-
i) Physical Level = How data is stored
ii) Logical Level = What data is stored
iii) View Level = what users sees.

Q4 = What is three schema architecture?

Ans = External level (user views)
Conceptual level (logical schema)
Internal level (physical storage)

Q5 = What is data independence?

Ans = Ability to modify schema at one level without affecting other levels.

Q6 = What is ER model?

Ans = Diagram that represents entities, attributes and relationships. used for database design.

Q7 = What are integrity constraints?

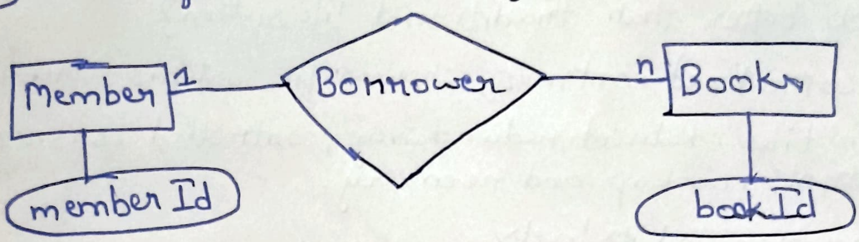
Ans = Rules that ensures correct and valid data.
Ex. primary, foriegn, unique, not null keys.

Section-B

1) Three schema architecture diagram

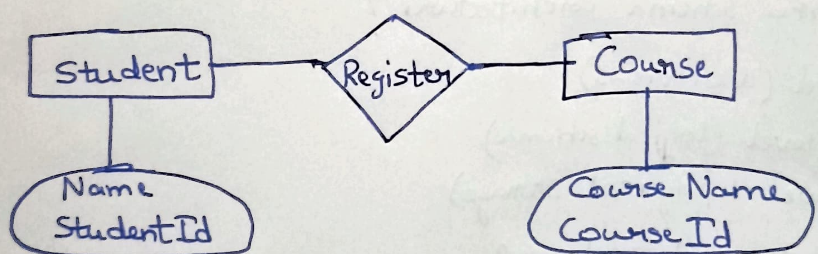
External level → view level
Conceptual level → logical structure
Internal level → Physical storage

2) i) ER diagram – Library



Entities → Member, Books
Relationship → Borrower
Attributes → memberId, BookId

ii) ER diagram – Student Course Registration



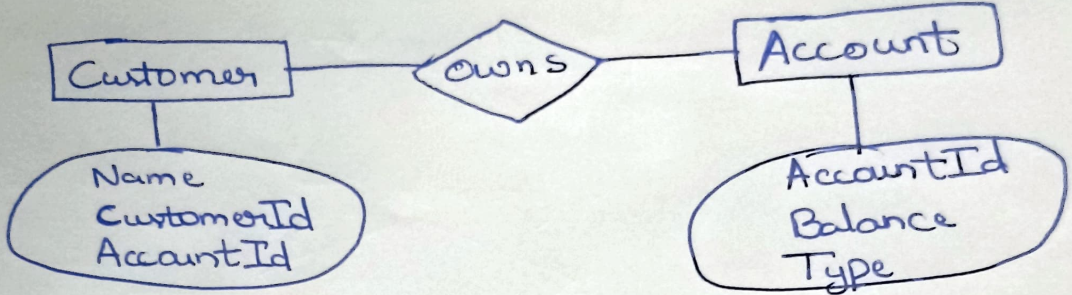
Section - C

1) ER diagram :- Hospital System
Entities → Patient, Doctor, Appointment, Patient books appointment, Doctor attends appointment
Primary key → PatientId, Doctor Id, Appoitment Id

2) Online Shopping
Customer (CustomerId - Pk)
Product (ProductId - Pk)
Order (OrderId - PK)
Foriegn key in order table connect customer and product

Section -D

1) ER diagram → ATM system



→ Account is primary key
→ Relation - customer owns account